

**DSC 40A - Homework 1**  
due Monday, July 6th at 11:59PM

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Homeworks are due to Gradescope by 11:59PM on the due date.

You may request an extension for this assignment by submitting [the form](#).

Homework will be evaluated not only on the correctness of your answers, but on your ability to present your ideas clearly and logically. You should **always explain and justify** your conclusions, using sound reasoning. Your goal should be to convince the reader of your assertions. If a question does not require explanation, it will be explicitly stated.

Homeworks should be written up and turned in by each student individually. You may talk to other students in the class about the problems and discuss solution strategies, but you should not share any written communication and you should not check answers with classmates. You can tell someone how to do a homework problem, but you cannot show them how to do it. **Only handwritten solutions will be accepted (use of tablets is permitted). Do not typeset your homework (using L<sup>A</sup>T<sub>E</sub>X or any other software).** Staff will not grade typed submissions.

For each problem you submit, you should **cite your sources** by including a list of names of other students with whom you discussed the problem. Instructors do not need to be cited.

This homework has 6 problems and will be graded out of 48 points. Good luck!

**Problem 1. Welcome Survey (2 points)**



Make sure to fill out the [Welcome Survey](#) for two points on this homework! I don't know about you, but I'd definitely take a couple of free points.

## Problem 2. Proofs (7 points)

In this problem, you'll prove or disprove various statements about a dataset of numbers,  $y_1, y_2, \dots, y_n$ . The goal for this problem is for you to practice rigor in writing mathematical proofs. As such, when making a statement, you must provide some sort of reason as to why it is valid. For instance, consider the statement:

*"Suppose we add 5 to each of  $y_1, y_2, \dots, y_n$ . The mean of the new dataset must be greater than the mean of the original dataset."*

This statement is always true, but it's not enough just to say "This statement is always true; since we're adding a positive number to each value, the mean will also increase." The intuition is good, but we need a more rigorous justification. For instance:

*"The mean of the original dataset is  $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$ . The mean of the new dataset is:*

$$\frac{1}{n} \sum_{i=1}^n (y_i + 5) = \frac{1}{n} \left( \sum_{i=1}^n y_i + \sum_{i=1}^n 5 \right) = \frac{1}{n} \left( \sum_{i=1}^n y_i + 5n \right) = \frac{1}{n} \sum_{i=1}^n y_i + 5 = \bar{y} + 5$$

*Therefore, the mean of the new dataset is equal to the original dataset's mean plus 5, so the mean of the new dataset is greater than the mean of the original dataset, and so the statement is always true."*

Make sure that your answers incorporate a similar level of rigor to show a deeper understanding.

Now, consider a dataset of numbers  $y_1, y_2, \dots, y_n$ . Prove or disprove each of the following statements.

- a) 🥑🥑 At least half of the numbers in the dataset must be smaller than the mean.

**Solution:**

- b) 🥑🥑 Suppose that all of the elements in the dataset are unique. Then, removing the largest element in the dataset must increase the mean.

**Solution:**

- c) 🥑🥑 Suppose that all of the elements in the dataset are unique, that  $n$  is odd, and that the mean of the dataset is not equal to the median of the dataset. Then, if we remove the median value from the dataset, the median of the new dataset must be different from the median of the original dataset.

**Solution:**

- d) 🥑 Suppose we introduce a new number to the dataset that is greater than the mean of the existing dataset. The mean of the new dataset must be greater than the mean of the original dataset.

**Solution:**

**Problem 3. Linear Functions (10 points)**

Consider the linear function  $f(x) = 3x - 2$ .

- a) 🥑 If  $a \leq b$ , show that  $f(a) \leq f(b)$ .

**Solution:**

- b) 🥑🥑 Both of the statements below are true, but only one is a consequence of the property you proved in part (a). Show which one it is and why using your results from part (a). Then, prove the other statement.

- (a)  $\text{Mean}(f(x_1), \dots, f(x_n)) = f(\text{Mean}(x_1, \dots, x_n))$   
(b)  $\text{Median}(f(x_1), \dots, f(x_n)) = f(\text{Median}(x_1, \dots, x_n))$

**Solution:**

- c) 🥑🥑 Now, prove the other statement.

**Solution:**

- d) Suppose we consider a different linear function  $g(x) = -2x + 4$ . Prove or find a counterexample to disprove each of the following:

- (a) 🥑 If  $a \leq b$ , then  $g(a) \leq g(b)$ .  
(b) 🥑🥑  $\text{Mean}(g(x_1), \dots, g(x_n)) = g(\text{Mean}(x_1, \dots, x_n))$   
(c) 🥑🥑  $\text{Median}(g(x_1), \dots, g(x_n)) = g(\text{Median}(x_1, \dots, x_n))$

**Solution:**

**Problem 4. Mean Absolute Error (6 points)**

In lecture, we found that  $h^* = \text{Median}(y_1, y_2, \dots, y_n)$  is the constant prediction that minimizes mean absolute error:

$$R_{\text{abs}}(h) = \frac{1}{n} \sum_{i=1}^n |y_i - h|$$

Your friend thinks that instead of minimizing the mean absolute error, we'd get a better constant prediction if we minimized the *product* of the absolute errors:

$$P_{\text{abs}}(h) = \prod_{i=1}^n |y_i - h|$$

In this problem, we'll see if your friend has a good idea.

- a) 🥑 Without using a calculator or computer, graph  $P_{\text{abs}}(h)$  for the dataset  $y_1 = -3, y_2 = 4$ .

**Solution:**

- b) 🥑🥑🥑 For an arbitrary data set  $y_1, y_2, \dots, y_n$ , what value(s)  $h^*$  minimize  $P_{\text{abs}}(h)$ ? Discuss the pros and cons of using your friend's prediction strategy. What factors about the data set or application will influence whether this prediction strategy gives good predictions?

**Solution:**

**Problem 5. Mean Squared Error (7 points)**

In lecture, we found that  $h^* = \text{Mean}(y_1, y_2, \dots, y_n)$  is the constant prediction that minimizes mean squared error:

$$R_{\text{sq}}(h) = \frac{1}{n} \sum_{i=1}^n (y_i - h)^2$$

To arrive at this result, we took the derivative of  $R_{\text{sq}}(h)$  with respect to  $h$ , set it equal to 0, and solved for the resulting value of  $h$ , which we called  $h^*$ .

In this problem, we will minimize  $R_{\text{sq}}(h)$  in a way that doesn't use calculus. The general idea is this: if  $f(x) = (x - c)^2 + k$ , then we know that  $f$  is a quadratic function that opens upwards with a vertex at  $(c, k)$ , meaning that  $x = c$  minimizes  $f$ . As we saw in class,  $R_{\text{sq}}(h)$  is a quadratic function of  $h$ .

Let  $y_1, y_2, \dots, y_n$  be an arbitrary dataset, and let  $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$  be the mean of each  $y_i$ .

- a) 🥑🥑 What is the value of  $\sum_{i=1}^n (y_i - \bar{y})$ ? Justify your answer.

**Solution:**

- b) 🥑🥑 Show that:

$$R_{\text{sq}}(h) = \frac{1}{n} \sum_{i=1}^n \left( (y_i - \bar{y})^2 + 2(y_i - \bar{y})(\bar{y} - h) + (\bar{y} - h)^2 \right)$$

*Hint: To proceed, start by rewriting  $y_i - h$  in the definition of  $R_{\text{sq}}(h)$  as  $(y_i - \bar{y}) + (\bar{y} - h)$ . Why is this a valid step? Make sure not to expand unnecessarily.*

**Solution:**

- c) 🥑🥑 Show that:

$$R_{\text{sq}}(h) = \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2 + (\bar{y} - h)^2$$

You will need to use your result from part (a) to solve part of this problem.

**Solution:**

- d) 🥑 Why does the result in (c) prove that  $h^* = \text{Mean}(y_1, y_2, \dots, y_n)$  minimizes  $R_{\text{sq}}(h)$ ?

**Solution:**

**Problem 6. New Loss Function (12 points)**

Suppose we are given a data set of size  $n$  with  $0 < y_1 \leq y_2 \leq \cdots \leq y_n$ .

Define a new loss function by

$$L_Q(h, y) = (h^2 - y^2)^2$$

and consider the empirical risk

$$R_Q(h) = \frac{1}{n} \sum_{i=1}^n L_Q(h, y_i).$$

- a) 🥑🥑🥑🥑 Show that  $R(h)$  has critical points at  $h = 0$  and when  $h$  equals the **quadratic mean** of the data, defined as

$$QM(y_1, y_2, \dots, y_n) = \sqrt{\frac{y_1^2 + y_2^2 + \cdots + y_n^2}{n}}.$$

- b) 🥑🥑🥑🥑 Recall from single-variable calculus the **second derivative test**, which says that for a function  $f$  with critical point at  $x^*$ ,

- if  $f''(x^*) > 0$ , then  $x^*$  is a local minimum, and
- if  $f''(x^*) < 0$ , then  $x^*$  is a local maximum.

Use the second derivative test to determine whether each critical point you found in part (a) is a maximum or minimum of  $R_Q(h)$ .

- c) 🥑🥑🥑🥑 Show that the quadratic mean always falls between the smallest and largest data values, which is a property that any reasonable prediction should have. This amounts to proving the inequality

$$y_1 \leq QM(y_1, y_2, \dots, y_n) \leq y_n.$$